Shiven Mehta

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Education

Georgia Institute of Technology | Atlanta, GA *August 2025 – Present*
GPA: 4.0/4.0
Bachelor of Science in Electrical Engineering, Minor in Computer Science (Intelligence) Expected Graduation, Dec. 2028

Technical Skills

- **Programming:** Python, Java, Javascript, Typescript HTML/CSS, C++, Matlab
- **Tools/Frameworks:** Git, React, NextJS, Source Tree, Chakra UI, OpenCV, Raspberry Pi, Arduino
- **Core Competencies:** Full-Stack Development, Frontend Development, Machine Learning, Data Analysis, API Development, Computer Vision, PCB/Embedded Systems
- **Languages**: English (Fluent), Gujarati (Conversational), Hindi (Conversational), Spanish (Beginner)

Experience

KonstructIQ | Los Altos, CA *May – July 2026*
Software Engineering Intern

- Developed and shipped 15+ Next.js/React features (TypeScript, Chakra UI) to KonstructIQ's platform, which is used by 5,000+ monthly users: covering change orders, daily logs, invoices, purchase orders, and project views; conducted testing and code review to ensure reliability across releases.
- Built LLM-powered agents using Apryse PDF SDK to extract structured tables from construction-plan PDFs and organize the results into structured outputs.
- Evaluated accuracy of KonstructIQ's AI estimator agent across dozens of real construction plans, identifying extraction discrepancies and informing prompt and workflow refinements.
- Collaborated on frontend/backend integration, permissions, error handling, and project data workflows while using Git.

Sport AI | Oslo, Norway (Remote) *June – August 2024*
Software Intern

- Annotated and corrected CVAT training datasets across 30+ hours of tennis/golf footage, reducing mislabeled instances and improving player-detection model accuracy.
- Co-developed RESTful Player-Detection API endpoints (TypeScript) for SportAI's ML team, enabling automated retrieval of model predictions for tennis and golf analytics.
- Enhanced company website by implementing SEO-optimized metadata and refactoring frontend code, improving article discoverability and load performance.

Bestronics - Emerald Technologies | San Jose, CA *June – August 2023*
Production Planning Intern

- Optimized SMT feeder configurations to ensure precise component placement on high-volume PCB assemblies, reducing setup inefficiencies for manufacturing lines.
- Managed component inventory and prepared 100+ work orders to support daily production schedules, improving throughput and reducing part-shortage delays.

Georgia Institute of Technology | Atlanta, GA *August 2026 – Present*
Undergraduate Teaching Assistant – ECE 2035: Programming for Hardware/Software Systems

- Selected to support students learning C/C++ programming, RISC-V assembly, debugging, memory organization, and hardware/software interfacing.
- Assist students during labs and office hours by explaining programming concepts, diagnosing implementation errors, and reinforcing systematic debugging practices.

Projects

Tennis Shot-Quality Prediction Model (Novel Model; 10-page Research Paper) *May – October 2024*

- Engineered a multi-model ML pipeline (TrackNet CNN, Faster R-CNN, CatBoostRegressor, LightGBM) to predict tennis shot quality (0–10 scale) from broadcast video, achieving a 7.31% MAPE and 0.44 MAE on a held-out test set.
- Built a mathematical hit-detection algorithm using binary moving averages on ball trajectory data, achieving ~90–95% shot-detection accuracy; used OpenCV for ball/player tracking and homography-based court calibration.
- Retrained a modified TrackNet court-detection model on 10,098 labeled frames across 5 matches with non-standard camera angles, improving detection robustness beyond the original pretrained model.
- Authored a 10-page research paper on the methodology and results; recognized with Honorable Mention and the ASEI Award at the Synopsys Science Fair.

Omnidirectional Robot with PID Motion Control | *St.Francis Science Olympiad* *November 2024 – March 2025*

- Led a team of 6 to design and build a mecanum-wheel robot using custom 3D-printed components, driven by stepper motors.
- Developed embedded control software in C++ (Arduino), integrating a quad motor controller and implementing a PID feedback loop using IMU data for precise alignment and motion accuracy.
- Achieved top-five finishes out of 50+ teams at the Stanford Invitational and NorCal Regionals.

Digital Design Lock | Personal Project *November 2025 – May 2026*

- Designed and simulated a fully hardware-implemented 2-digit digital lock using FSMs, D latches, JK flip-flops, and 4-bit register storage, validated in the Lush Project circuit simulator.
- Implemented display and control logic using a 7-segment decoder and designed an 8-bit gate-level comparator, integrated with an FSM to support password verification, reset, and reconfiguration modes.

ClimaQuant | Full-Stack Climate-Adjusted Options Pricing Platform *March – April 2026*

- Built a full-stack web application integrating NOAA climate indicators (HDD/CDD, Palmer Z Drought Index) with equity data to model climate-adjusted volatility in Black–Scholes option pricing.
- Built an interactive dashboard for company option pricing simulations, allowing dynamic parameter tuning (strike price, maturity, risk-free rate) and performance evaluation via Absolute Average Percent Error (AAPE).

Georgia Tech Formula SAE Keychain Project *September – October 2025*

- Created the schematic, 3D footprint, and PCB layout in Altium for a custom USB-C powered keychain featuring an L555 timer circuit controlling six LEDs that replicate the car's "Ready to Move" signal.
- Completed PCB design training in Altium; currently undergoing firmware training in C++.

Personal Website *March – April 2026*

- Independently designed, styled, and developed a personal portfolio website (HTML/SCSS/JS) to showcase engineering projects, writing custom CSS architecture with SCSS partials/variables rather than a framework or template.

Full Stack Tennis Booking Platform *May – July 2026*

- Built and deployed a full stack booking web application with a custom interactive calendar, real-time slot availability, and automated email confirmations with Google Calendar invites, facilitating 20+ tennis sessions with local players in Sunnyvale, CA
- Implemented a RESTful Express.js backend with PostgreSQL database, server-side validation, rate limiting, and input sanitization, deployed on Heroku with a Next.js frontend on Netlify.

Stock Portfolio Tree - React, TypeScript, Chakra UI, Vite *August 2026 – Present*

- Built an interactive React/TypeScript tool for visualizing investment portfolios as editable node-link trees, supporting multiple saved portfolios with persistent state.
- Implemented a custom hierarchical layout algorithm from scratch (bottom-up subtree-width computation with top-down position assignment) to guarantee non-overlapping branches for arbitrary node counts.
- Integrated the Finnhub API with debounced ticker search and real-time price lookups, minimizing redundant network calls during user input.

Bridge — Budget-Conscious Meal Planning | HackGT 2026

Next.js, TypeScript, Python, FastAPI, Supabase, Vercel, Grok API

- Built and deployed a full-stack application generating personalized weekly meal plans based on budget, dietary restrictions, cuisine preferences, and nutrition goals.
- Implemented a Python KNN ranker comparing five nutrients across portion sizes, with AI-assisted scheduling and deterministic checks for eligibility, meal variety, and household scaling.
- Integrated 313 MealDB recipes, Grok nutrition estimates, and location-based Kroger pricing; reduced pricing requests by shortlisting the top 25 meals per category and caching ingredient prices.
- Developed consolidated grocery lists and community food-access tools, with persistent planning workflows and private plan storage in Supabase.

Diamond Path Optimizer | C, RISC-V Assembly *March – April 2026*

- Developed a constrained graph-search algorithm in C and RISC-V assembly to compute the minimum-cost five-step path from the center of an 11×11 weighted diamond grid to any edge.
- Optimized the assembly implementation using level-based dynamic programming, quadrant symmetry, reusable subroutines, and register reuse to reduce instruction count, execution time, and memory requirements.

Relevant Coursework

Digital System Design: Combinational/Sequential Logic Circuit Design, RISC-V Architecture, Circuit Timing, CMOS System Design
Programming for Hardware/Software Systems: Embedded C/C++ development, RISC-V Assembly Language, Memory-mapped I/O
Data Structures & Algorithms: Fundamental data structures (arrays, linked-lists, stacks, queues, trees, hash maps), and Algorithm design techniques (recursion, sorting, searching, big-o notation)
Circuit Analysis: Op-amps, RLC circuits, low & high-pass filters, AC signals/power
Signal Processing: AM/FM signals, Signal Analysis in Matlab, ADC & DAC converters